

# UAM/ AAM

1. AAM - Research Direction CNS ( Kamesh Namuduri , Uwe-Carsten Fiebig , David W. Matolak , Ismail Guvenc , K.V.S. Hari, and Helka-Liina Määttänen)

- Readiness level of CNS in AAM
- A2A communication , 3GPP Support for Nav and DAA
- A key subsystem in the AAM architecture [2] is the provider of services for UAM (PSU), which will provide services to support operations planning, flight intent sharing, strategic and tactical deconfliction, and airspace management functions. PSUs will exchange information with other PSUs via a network that enables safe, efficient operation within the AAM corridors—volumes of airspace dedicated to AAM use.
- It also ensures discovery and synchronization of services and include supplementary data service providers, which provide services such as weather information, and a flight management information system
- ASTM on Data exchange
- most critical CNS challenges to which academic researchers can contribute. These include:
  - air corridors
  - AA communications
  - 3GPP support for navigation
  - DAA/collision avoidance.
- navigation in GPS-denied environments, noise mitigation, and security and privacy or other topics
- Air corridors are 3D volumes of airspace reserved for UASs for AAM traffic.
- In the United States, the Federal Aviation Administration (FAA) defines air corridors in class B, C, or D airspaces.
- V2V is presumed as automobile to automobile so use AA/ A2A

- Standardization efforts for AA communication protocols, use cases, spectrum needs, and security requirements are currently taking place in the AAM community.
- AA communications can be combined with the principles of radar systems to develop **dual-function radar communication (DFRC) systems** [4] where radar waveforms can be designed to carry information to communicate with other vehicles in the vicinity. The dual use of DFRC for sensing and communications has potential applications in AAM, particularly in the CNS domain.
- with terrestrial settings, multiple network topologies (e.g., mesh, relay, star, and so on) can be employed.
- AA communications will likely serve a supporting role to primary AG communication links, whereas navigation, and, in particular, surveillance, will rely more on AA signaling.
- For all three functions (CNS), reliability at the lower layers of the communications protocol stack is critical.
- AA signaling at the physical and data link layers.
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- When link distances are on the order of altitudes or smaller, the AA channel is very close to a free-space channel, assuming mostly azimuth covering antennas. This is easily deduced from geometry and basic propagation principles. The primary obstacle that affects the AA channel is the Earth surface, and, as link distances increase, reflections from the Earth surface become the next significant channel component in addition to the line-of-sight (LOS) component [5]. For lower aircraft altitudes, the AA link range will decrease since the radio LOS over the curved Earth is proportional to the square root of the altitude. In addition, the probability of obstruction of the AA link increases as the altitude decreases.

- For links between aircraft that are not too close to the ground and are at approximately the same altitude, this  $1/d^2$  free-space path loss model allows large link ranges in comparison to terrestrial settings, where, often, a  $1/d^3$  or  $1/d^4$  attenuation relation holds.
- At altitudes low enough to allow significant reflections from terrestrial obstacles, additional multipath components (MPCs) may be present. These can occur at relatively large values of delay and, with aircraft motion, will tend to be intermittent, as found for the AG channel [7].
- Directional antennas suppress MPCs and can enable larger link ranges, but, in a dynamic AA setting, antenna pointing and tracking can be challenging. For AAM applications, some of the most challenging cases will be in the near-urban and urban environments. In the near urban case, flights will likely follow established air corridors well above buildings; hence, AA channels may be well modeled by a two-ray channel, in which the terrain cover type will determine the strength and nature of the reflection (specular or diffuse). Work on estimating and validating these air corridor channel models should be done for the appropriate frequency bands.
- For the urban case, additional MPCs will be present from buildings and other terrestrial structures (e.g., water tanks or highway overpasses) that will also sometimes act as obstructions. Obstruction attenuation generally increases with frequency. To date, largely because of the difficulty of conducting flight tests over and within urban areas, most channel models for this case rely solely on analyses and simulations. Thus, actual experimental work on this topic would be very valuable.
- A first wideband measurement campaign, which aims at getting a deeper understanding of signal propagation in urban AA environments, was conducted in 2019 [8].
- Read the test experiment setup for AA propagation condition analysis, LOS as well as non-LOS situations have been assessed.
- Those scenarios are crucial for DAA since they may either interrupt the AA communication link or result in an incorrect reception of messages.

- AA links will employ the same signaling schemes as AG links. AA signalling based on application
- very high frequency digital signal , Link 16 and Common Data Link
- From 3GPP's perspective, both ( C2 and UAS application data) categories are "user-plane" traffic.
- C2 - Latency and Biterror rate were concluded more important than data rate
- The outcome of the study was documented in the 3GPP technical report [9], which also included field measurement results. In [10], the authors present a related study and findings of UAS operation in cellular networks.
- Radio Resource Control (RRC) , Broadcast ID, Device ID
- signal strength measured at the UAS when N neighboring base stations' (BSs') signal strengths are above a threshold.
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- reporting of the signal strength measured at the UAS when N neighboring base stations' (BSs') signal strengths are above a threshold. This, together with the height reporting, is used for both interference detection and flying mode detection. One key feature is core network signaling for subscription-based identification for the network to verify whether or not a mobile device can be served as a UAS in the network.
- signaling flight path information from the UAS to the BS was added to the air interface signaling protocol. This includes network polling of list of waypoints (3D locations) and time stamps if available. These enhancements are described in more detail in chapter 6 in [12].
- 3GPP enables a D2D link, also defined as a sidelink or PC5, (cellular intelligent transport system
- This provides a framework for supporting UAS remote ID broadcasting. As an LTE BS may schedule 5G NR user equipment (UE), and 5G NR BS may schedule LTE

UE, it has been agreed in release 18 to study and specify if the support for this feature is needed for both NR and LTE.

- release 18 discussions and is proposed to be addressed during the 3GPP normative work.
- The PC5 already supports broadcasting messages for collision avoidance over a sidelink, and the reuse of these traffic-safety messages should be the starting point.
- The release 18 UAS discussions also include considering further enhancements for uplink beamforming and beam management to further control the interference and to improve the link quality with faster switching of uplink beams toward the BS. However, these suggestions need to be considered against the multiple-input, multiple-output release 17 enhancements, which may already provide the needed level of improvements.
- **Conditional Handover (HO with condition)**
- Currently, the conditional HO command may include up to eight target cells and up to two conditions per target cell, and the conditions can be signal strength related only.
- the required information over a communication link, be it AA communication or AG communication, as well as from local sensors, such as cameras, radars, and lidars. The communication messages contain information about the current position, current velocity vector, future waypoints, destination, and priority mode as well as about the size, volume, and freight type of a particular UAS.
- remote ID-capable drone option, which is one of the three modes) that the drone broadcasts some critical information. Such information includes the drone's unique ID, location and altitude, velocity, control station location and elevation, time mark, and emergency status.
- spectrum similar to what is used by Wi-Fi and Bluetooth, while the rule making does not specify or mandate a fixed frequency or technology. Such remote ID information, if it can be reliably, seamlessly, and continuously monitored, will

be the primary source for

DAA operations. Additional research is needed to compare the suitability of different wireless technologies for remote ID operations.

- The RF signals between a drone and its controller can also be passively monitored for detecting, classifying, localizing, and tracking a drone [15].
- Local sensors, like cameras, can be used complementarily to communication links but require good visibility conditions, a high visual resolution, and sufficient onboard computing power. Also, radar and lidar can complement or back up communication links.
- When the DAA mechanisms identify that there is a collision course involving one or more UASs, centralized (the UAS action is communicated by a remote entity) or UAS-centric (each UAS takes its own decision) measures can be taken. When there are collision courses where more than two UASs are involved, there

is a need to develop novel avoid concepts. These situations occur when UASs fly on air corridors one after the other: when an unmanned aerial vehicle (UAV) has to decelerate, it has an impact on all other UASs following the first one—similar to platooning on roads for self driving vehicles. Correction courses may use the third dimension, causing UASs to use air layers where no UAS is expected to fly. However, if, in these air layers, other UASs are already present, new conflicts may arise. With only a few UASs in operation, this issue is not of great significance, but it will become crucial with a growing number of UASs. When collision avoidance maneuvers cause further conflicts, the overall stability of AAM is in danger.